EE1101 Signals and Systems APR-JUN 2021
Tutorial 4 Solutions

1) For causal Discrete time LTI systems, h[n]=0 for n<0
For stable Discrete time LTI systems, $\sum_{n=-\infty}^{\infty} |h[n]| < \infty$.

For Causal Continuous time LTI systems, h(t)=0 for t<0

For Stable Continuous time LTI systems, $\int_{\tau=-\infty}^{\infty} |h(\tau)| d\tau < \infty$.

(a) h[n] = $(-\frac{1}{2})^n u[n] + (1.01)^n u[1-n]$
Since u[1-n]$\neq$ 0 for n< 0 , substituting $n < 0$ gives h[n] $\neq 0$
Therefore ,the system is NON CAUSAL
$\sum_{n=-\infty}^{\infty} |h[n]| = \sum_{n=-\infty}^{\infty} |(-\frac{1}{2})^n u[n] + (1.01)^n u[1-n]|$
$\sum_{n=-\infty}^{\infty} |(-\frac{1}{2})^n u[n]| = \sum_{n=0}^{\infty} |(-\frac{1}{2})^n| < \infty$
$\sum_{n=-\infty}^{\infty} |(1.01)^n u[1-n]| = \sum_{n=-\infty}^{1} |(1.01)^n| < \infty$
Therefore, $\sum_{n=-\infty}^{\infty} |h[n]| < \infty$. So the system is STABLE

(b) h[n] = $n(\frac{1}{3})^n u[n-1]$
Since u[n-1]= 0 for n< 1, h[n] = 0 for n<0
Therefore ,the system is CAUSAL

Intuitively, $n(\frac{1}{3})^n u[n-1]$ has a converging sum.Therefore, $\sum_{n=-\infty}^{\infty} |h[n]| < \infty$.
To prove mathematically,

$$\sum_{n-\infty}^{\infty} |h[n]| = \sum_{n=-\infty}^{\infty} |n(\frac{1}{3})^n u[n-1]|$$
$$= \sum_{n=1}^{\infty} |n(\frac{1}{3})^n u[n-1]|$$
$$= \frac{1}{3} + \frac{2}{3^2} + \frac{3}{3^3} + ...(sum\ of\ infinite\ AGP)$$
$$= 0.75 < \infty$$

Therefore,the system is STABLE

(c) h(t) = $e^{2t} u(-1-t)$
Since u(-1-t)$\neq$ 0 for t< 0, h(t))$\neq$ 0 for t <0;
Therefore,system is NON CAUSAL

$$\int_{\tau=-\infty}^{\infty} |h(\tau)|d\tau = \int_{\tau=-\infty}^{\infty} |e^{2\tau}u(-1-\tau)|d\tau$$
$$= \int_{\tau=-\infty}^{-1} |e^{2\tau}|d\tau$$
$$= \frac{e^{-2}}{2} < \infty$$

Therefore,the system is STABLE

(d) h(t) = $e^{-6|t|}$
Since h(t) has values at all instants of time, h(t))$\neq$ 0 for t <0;
Therefore,system is NON CAUSAL

$$\int_{\tau=-\infty}^{\infty} |h(\tau)|d\tau = \int_{\tau=-\infty}^{\infty} |e^{-6|\tau|}|d\tau$$
$$= \int_{\tau=-\infty}^{0} |e^{6\tau}|d\tau + \int_{\tau=0}^{\infty} |e^{-6\tau}|d\tau$$
$$= |\frac{1}{6}| + |\frac{1}{6}| = \frac{1}{3} < \infty$$

Therefore,the system is STABLE

(e) h(t) = $(2e^{-t} - e^{(t-100)/100})u(t)$
Since h(t) is some function multiplied by u(t), h(t))= 0 for t <0;
Therefore,system is CAUSAL

$$\int_{\tau=-\infty}^{\infty} |h(\tau)|d\tau = \int_{\tau=-\infty}^{\infty} |(2e^{-\tau} - e^{(\tau-100)/100})u(\tau)|d\tau$$
$$= \int_{\tau=0}^{\infty} |(2e^{-\tau} - e^{(\tau-100)/100})|d\tau$$

In this, We have an exponentially decaying part ($(2e^{-\tau}$),and
Exponentially increasing part ($e^{(\tau-100)/100}$)
The exponentially increasing part makes the integral sum to infinte. Therefore ,system is UNSTABLE

2)

(a) $h(t) = \cos(\pi t)$
(i) This system is not memoryless, since $h(t)$ is not zero for $t \neq 0$.
(ii) This system is not causal, since $h(t)$ is not zero for $t < 0$.
(iii) This system is not stable, since $\int_{-\infty}^{\infty} |h(t)|dt = \infty$ (i.e., it is not absolutely integrable).

(b) $h(t) = e^{-2t}u(t-1)$
(i) This system is not memoryless, since $h(t)$ is not zero for $t \neq 0$.
(ii) This system is causal, since $h(t)$ is zero for $t \leq 0$.
(iii) This system is stable, since $\int_{-\infty}^{\infty} |h(t)|dt = e^{-2}/2$.

(c) $h(t) = u(t+1)$
(i) This system is not memoryless, since $h(t)$ is not zero for $t \neq 0$.
(ii) This system is not causal, since $h(t)$ is not zero for $t < 0$.
(iii) This system is not stable, since $\int_{-\infty}^{\infty} |h(t)|dt = \infty$.

(d) $h(t) = 3\delta(t)$
(i) This system is memoryless, since $h(t)$ is zero for $t \neq 0$.
(ii) This system is causal, since it is memoryless.
(iii) This system is stable, since $\int_{-\infty}^{\infty} |h(t)|dt = 3$.

(e) $h(t) = \cos(\pi t)u(t)$
(i) This system is not memoryless, since $h(t)$ is not zero for $t \neq 0$.
(ii) This system is causal, since $h(t)$ is zero for $t < 0$.
(iii) This system is not stable, since $\int_{-\infty}^{\infty} |h(t)|dt = \infty$.

(f) $h[n] = (-1)^n u[-n]$
(i) The system has memory since $h[n] \neq 0$ whenever $n \neq 0$.
(ii) Since $h[n] \neq 0$ when n is negative, the system is noncausal.
(iii) Since $\sum_{n=-\infty}^{\infty} |h[n]|$ is infinite and the system is unstable.

(g) $h[n] = (1/2)^{|n|}$
(i) The system has memory since $h[n] \neq 0$ whenever $n \neq 0$.
(ii) Since $h[n] \neq 0$ when n is negative, the system is noncausal.
(iii) Since $\sum_{n=-\infty}^{\infty} |h[n]|$ is finite, the system is stable.

(h) $h[n] = \cos(\pi n/8)\{u[n] - u[n-10]\}$
(i) The system has memory since $h[1] \neq 0$.
(ii) Since $h[n] = 0$ when n is negative, the system is causal.
(iii) Since $\sum_{n=-\infty}^{\infty} |h[n]|$ is finite (since this is a finite sum, i.e., there is a finite number of terms in the summation), the system is stable.

(i) $h[n] = 2u[n] - 2u[n-5]$
(i) The system has memory since $h[1] \neq 0$.
(ii) Since $h[n] = 0$ when n is negative, the system is causal.
(iii) Since $\sum_{n=-\infty}^{\infty} |h[n]|$ is finite (since this is a finite sum, i.e., finite number of terms in the summation), the system is stable.

(j) $h[n] = sin(\pi n/2)$
(i) The system has memory since $h[n] \neq 0$ whenever $n \neq 0$.
(ii) Since $h[n] \neq 0$ when n is negative, the system is not causal.
(iii) Since $\sum_{n=-\infty}^{\infty} |h[n]|$ is infinite, the system is unstable.

(k) $h[n] = \sum_{p=-1}^{\infty} \delta[n-2p]$
(i) The system has memory since $h[-2] = 1 \neq 0$.
(ii) Since $h[-2] \neq 0$, the system is noncausal.
(iii) Since $\sum_{n=-\infty}^{\infty} |h[n]| = \infty$, the system is unstable.

3)

(a) (a) h[n] = $(-1/2)^n$u[n]

for n< 0

$$s[n] = 0$$

for $n \geq 0$

$$\text{s[n]} = \sum_{k=0}^{n} (\tfrac{-1}{2})^k$$

Using G.P. formula for finite series

$$\text{s[n]} = \tfrac{1}{3}(2 + (\tfrac{-1}{2})^n)u[n]$$

(b) h[n] = nu[n]

for $n < 0$

$$s[n] = 0$$

for n$\geq$ 0

$$s[n] = \sum_{k=0}^{n} k$$

$$\text{s[n]} = \tfrac{n(n+1)}{2} u[n]$$

(c) h(t) = $e^{-|t|}$

for $t < 0$

$$\text{s(t)} = \int_{-\infty}^{t} e^{\tau} d\tau = e^t$$

for t$\geq$ 0

$$\text{s(t)} = \int_{-\infty}^{0} e^{\tau} d\tau + \int_{0}^{t} e^{-\tau} d\tau = 2\text{- } e^{-\tau}$$

$$\text{s(t)} = \begin{cases} e^t & t < 0 \\ 2 - e^{-t} & t \geq 0 \end{cases}$$

(d) h(t)=(1/4)(u(t)-u(t-4))

for $t < 0$

$$\text{s(t)} = 0$$

for $t < 4$

$$\text{s(t)} = \tfrac{1}{4}\int_{0}^{t} d\tau = \tfrac{1}{4}t$$

for t$\geq$ 4

$$\text{s(t)} = \tfrac{1}{4}\int_{0}^{4} d\tau = 1$$

$$\text{s(t)} = \begin{cases} 0 & t < 0 \\ \frac{1}{4}t & 0 \leq t < 4 \\ 1 & t \geq 4 \end{cases}$$

(e) h(t) = u(t)

for $t < 0$

$$s(t) = 0$$

for t$\geq$ 0

$$s(t) = \int_0^t d\tau = t$$

$$s(t) = tu(t)$$

4)

(a) True. If $h(t)$ periodic and nonzero, then

$$\int_{-\infty}^{\infty} |h(t)|dt = \sum_{n=-\infty}^{+\infty} \int_{(n-1)T}^{nT} |h(t)|dt.$$

Since each summand is same , the infinite sum is unbounded. Thus $h(t)$ is unstable.

(b) False. For instance, suppose that the inverse of $h[n] = \delta[n - n_0]$ is $g[n]$. Then,

$$\Rightarrow h[n] * g[n] = \delta[n]$$
$$\Rightarrow \sum_{k=-\infty}^{\infty} \delta[k - n_0]g[n - k] = g[n - n_0] = \delta[n]$$
$$\Rightarrow g[n] = \delta[n + n_0]$$

which is noncausal.

(c) False. For example $h[n] = u[n]$ implies that

$$\sum_{-\infty}^{\infty} |h[n]| = \infty.$$

This is an unstable system.

(d) True. Assuming that $h[n]$ is bounded in the range $n_1 \leq n \leq n_2$,

$$\sum_{k=n_1}^{n_2} |h[k]| < \infty.$$

This implies that the system is stable.

(e) False. For example, $h(t) = tu(t)$ is causal but not stable.

(f) False. For example, the cascade of a causal system with impulse response $h_1[n] = \delta[n - 1]$ and a non-causal system with impulse response $h_2[n] = \delta[n + 1]$ leads to a system with overall impluse response given by $h[n] = h_1[n] * h_2[n] = \delta[n]$.

(g) False. For example, if $h(t) = e^{-t}u(t)$, then $s(t) = e^{-t}u(t) * u(t) = \int_{\tau=-\infty}^{\infty} e^{-\tau}u(\tau)u(t - \tau)d\tau$
$= \int_{\tau=0}^{t} e^{-\tau}u(\tau)d\tau = (1 - e^{-t})u(t)$ and

$$\int_0^{\infty} |1 - e^{-t}|dt = t + e^{-t}|_0^{\infty} = \infty.$$

Although the system is stable, the step response is not absolutely integrable.

(h) True. We may write $u[n] = \Sigma_{k=0}^{\infty}\delta[n-k]$. Therefore,

$$s[n] = \Sigma_{k=-\infty}^{\infty}u[k]h[n-k] = \Sigma_{k=0}^{\infty}h[n-k].$$

If $s[n] = 0$ for $n < 0$, then $h[n] = 0$ for $n < 0$ and the system is causal.

5) Given: System A is LTI and system B is inverse of A. Let $y_1(t)$ and $y_2(t)$ be outputs of system A for inputs $x_1(t)$ and $x_2(t)$ respectively. Combining these informations, we get,

$$x_1(t) \xrightarrow{\text{A}} y_1(t) \xrightarrow{\text{B}} x_1(t). \tag{1}$$

And,

$$x_2(t) \xrightarrow{\text{A}} y_2(t) \xrightarrow{\text{B}} x_2(t). \tag{2}$$

(a) **To prove system B is linear**. Assume that system B is not linear. From equations 1 and 2, we observe that an input $ax_1(t) + bx_2(t)$ to system A can generate $ay_1(t) + by_2(t)$, i.e.,

$$ax_1(t) + bx_2(t) \xrightarrow{\text{A}} ay_1(t) + by_2(t).$$

This is due to linearity property of the system A. Our assumption that system B is not linear implies that the output of B for the input $ay_1(t) + by_2(t)$ is not $ax_1(t) + bx_2(t)$, i.e. the outputs of system B does not add up linearly even if the inputs combine linearly. Therefore, we arrive at a situation which is as follows:

$$ax_1(t) + bx_2(t) \xrightarrow{\text{A}} ay_1(t) + by_2(t) \not\xrightarrow{\text{B}} ax_1(t) + bx_2(t).$$

This contradicts the fact that B is inverse of A. Hence, our assumption is incorrect, and so, system B is linear.

(b) **To prove system B is time-invariant**. Assume system B is time variant. By time-invariant property of system A, we have,

$$x_1(t-\tau) \xrightarrow{\text{A}} y_1(t-\tau).$$

When this output of system A, is fed to system B, we must not expect its response to be $x_1(t-\tau)$ because of our assumption. So, we land up in a situation where,

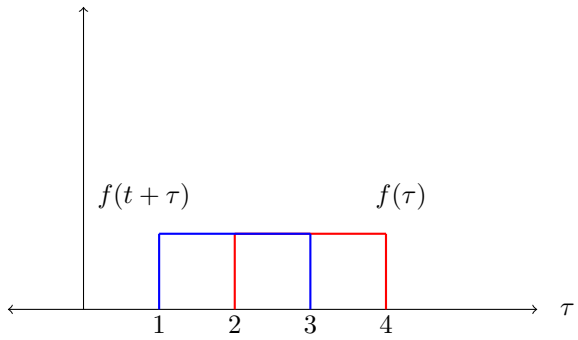
$$x_1(t-\tau) \xrightarrow{\text{A}} y_1(t-\tau) \not\xrightarrow{\text{B}} x_1(t-\tau).$$

This contradicts the fact that B is the inverse of A. So, the assumption about system B is incorrect. Therefore, system B is also time-invariant.

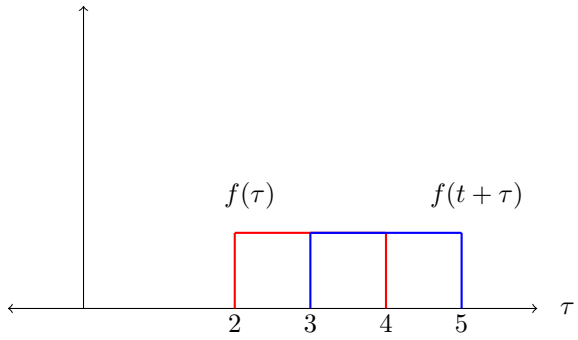
(6)

(a) $y[n]$ is $x[n]$ convolved with itself. $x[n]$ is a rectangular pulse with amplitude 1 in the range $2 \leq t \leq 4$. We know that, a rectangular pulse when convolved with itself, will yield a triangular signal. range of the signal t=(2+2) to t=(4+4). peak is at t=6 with a peak value (4-2).

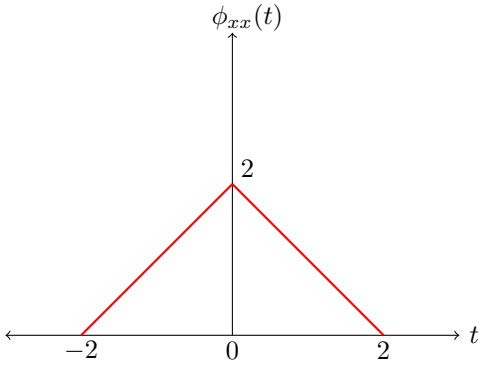
(b) autocorrelation of the given signal. for $t > 0$, say t=1



for $t < 0$, say t=-1



so the output is



7)

(a) Conjugate symmetric part of $x[n], x_e[n] = 0.5(x[n] + x^*[-n]$. Because $x_e[n]$ is conjugate symmetric, its real part is even and imaginary part is odd.

(b) $x[n] = je^{j4\pi n}$. So $x^*[-n]$ is $-je^{j4\pi n}$. Therefore conjugate symmetric part of the signal is zero. Signal is conjugate antisymmetric.

8)

(a) $y[n] = 2x[n]$ System is invertible. We can find x[n] for all n, when y[n] is divided by2.

(b) $y[n] = nx[n]$ System is not inverible as it is not possible to recover the input at n=0. Whatever may be the value of $x[0]$, all gets mapped to zero.

(c) $y[n] = x[n] - x[n-1]$ System is not invertible. Two constant signals will get mapped to the same output $y[n] = 0$.

(d) $x[n] = Rex[n]$ System is not invertible, as it is not possible to recover the imaginary part.

9) Given system: $y[n] = ay[n-1] + x[n]$, Input: $K\delta[n]$, Auxillary condition: $y[-1] = 3$.

(a) Find the expression for $y[n]$: **[(For $n \geq 0$)]** $y[n] = ay[n-1] + x[n]$. So, $y[0] = 3a + K, y[1] = a(3a + K) = 3a^2 + K, y[2] = 3a^3 + a^2K.....y[n] = 3a^{n+1} + a^nK$. **[For $(n < 0$)]** $y[n-1] = a^{-1}(y[n] + x[n])$. So, $y[-1] = 3, y[-2] = 3a^{-1}, y[-3] = 3a^{-2}.....y[n] = 3a^{n+1}$ So, $y[n] = 3a^{n+1} + Ka^nu[n]$ **for all n**

(b) System is not linear as the output is not zero when the input is zero. For this set K = 0, which means zero input. The resulting $y[n] = 3a^{n+1}$ and is non zero. System is Time variant. Can be checked by taking $x[n] = K[n - n_o]$. Quick check: This does not shift the output values for $n < 0$. So the system is not LTI.